SE311 SPRING 2025-2026 / 01-04-2026 & 02-04-2026
Lab Work 06 – Façade and Singleton Pattern

Look at the example code OfficeStuff.java in your Blackboard folder. Try to understand the key idea behind it.

Sisyphos arrives at the office and decides to start an intensive deep-work coding session. To prepare for the perfect environment, several smart devices in the office need to work together. First, he powers the workstation and launches his preferred development IDE. Then, he adjusts the climate control system to a professional and constant temperature. After that, he prepares himself a double espresso to drink it. He also opens the smart blinds to see the sunlight. Finally, he activates the DND lights to focus more on his room.

1. Create a Façade class called "**`OfficeAssistantSystem`**". This class will connect all those sub-system operations into one interface.

2. Create a method called **`startDeepWork(String ideName)`** that method will do the corresponding actions that exist in the previous code structure in **`OfficeAssistantSystem.`**

3. Make that **`OfficeAssistantSystem`** class Singleton with at most two instances using eager instantiation. Return the instance randomly chosen.

What must be done to make the **`OfficeAssistantSystem`** Singleton class Thread SAFE?

FOR EVERY STEP MAKE THE NECESSARY CHANGES ON THE EXISTING CODE!